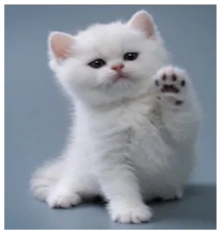
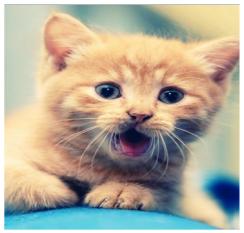
CMC



Kitten A



Kitten B

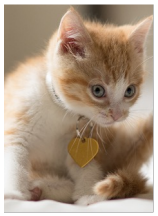


Kitten C

And four test queries:



1



2



3



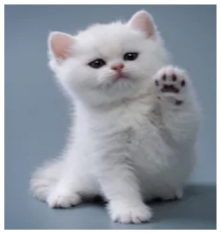
4

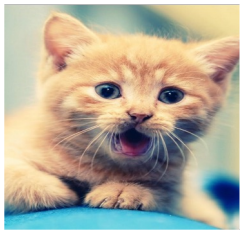
Query	Top result	Result 2	Result 3
1	A	B	C
2	B	C	A
3	A	B	C
4	C	B	A

What are the rank accuracies?

Recap: CMC

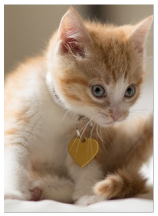

Kitten A


Kitten B


Kitten C

And four test queries:


1


2


3

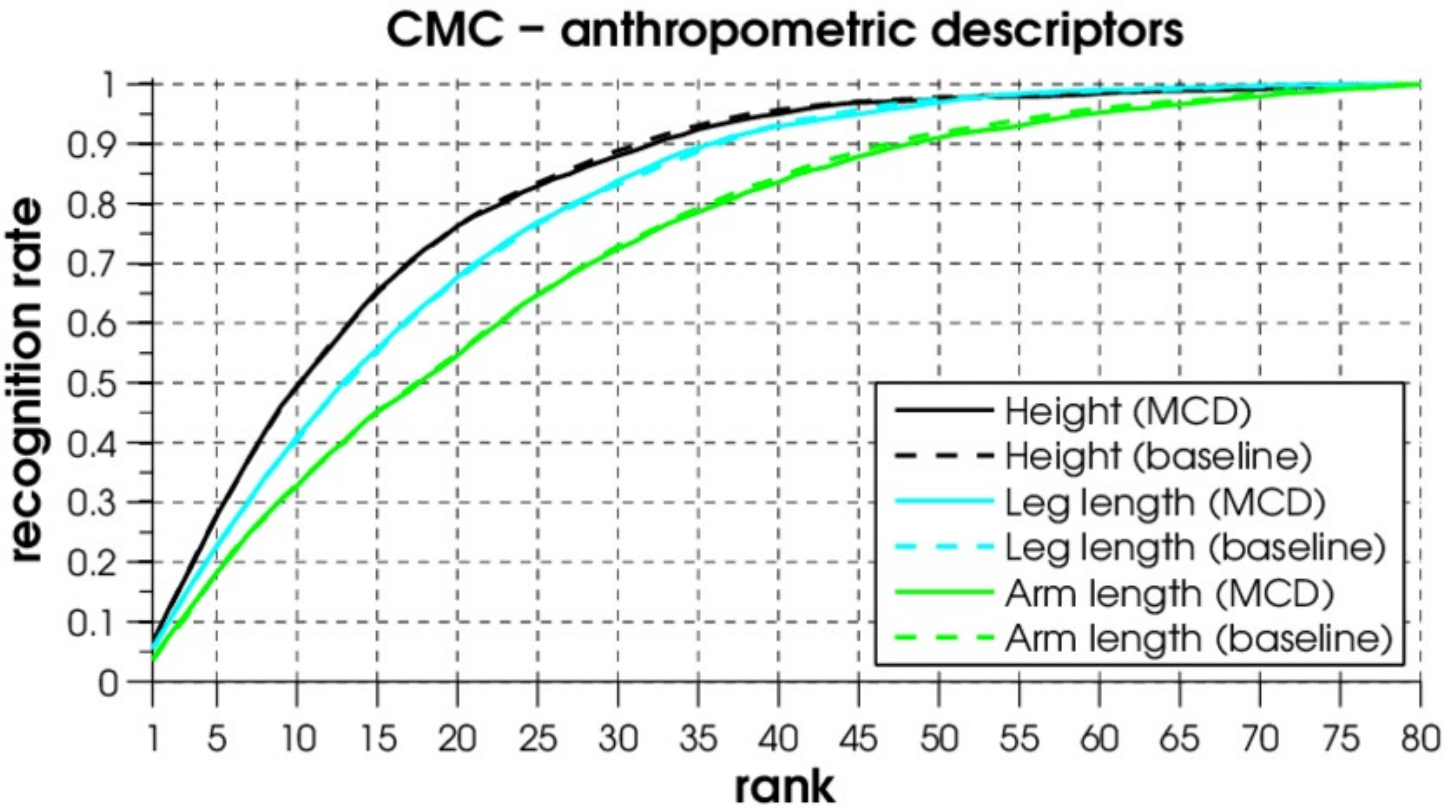

4

Query	Top result	Result 2	Result 3
1	A	B	C
2	B	C	A
3	A	B	C
4	C	B	A

What are the rank accuracies?

Rank 1: 1/4 predicted correctly: 25%
Rank 2: 3/4 : 75%
Rank 3: 4/4 : 100%

CMC Curve

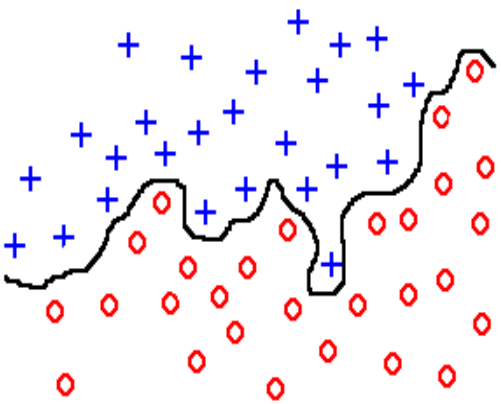

CMC – anthropometric descriptors
recognition rate
rank
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
1
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
Height (MCD)
Height (baseline)
Leg length (MCD)
Leg length (baseline)
Arm length (MCD)
Arm length (baseline)

Evaluating ML Systems

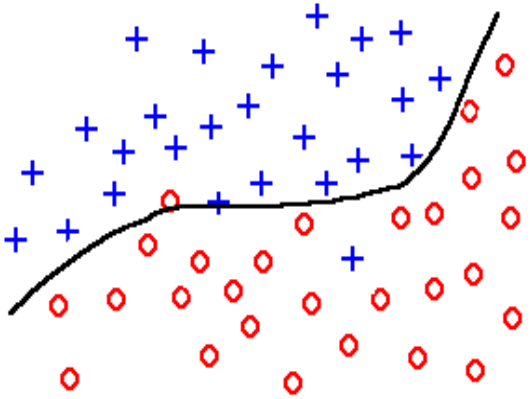
- Assumption: Building a model for population
- Reality: Population is not available
- We work with a sample database – not necessarily true representation of the population

- What to do?
  - Should we use the entire available database for training the model?
    - High accuracy on the training data
    - Lower accuracy on the testing data
    - Called as overfitting

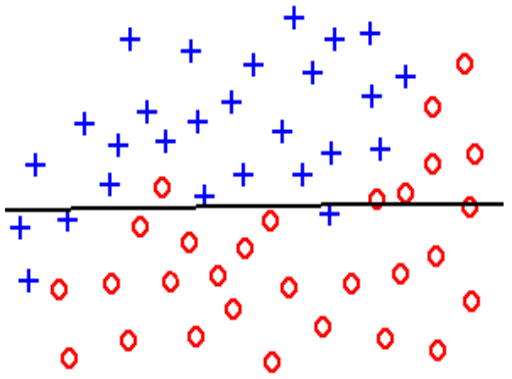
Evaluating ML Systems



Overfitting



Good fit



Underfitting

- Underfitting: Learning algorithm had the opportunity to learn more from training data, but didn't
- Overfitting: Learning algorithm paid too much attention to idiosyncracies of the training data; the resulting tree doesn't generalize

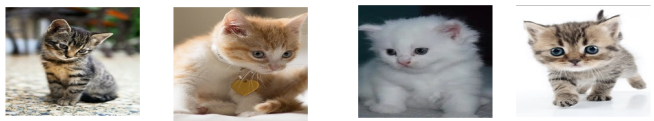
Cross Validation

- "Cross-Validation is a statistical method of evaluating and comparing learning algorithms."
- The data is divided into two parts:
  - Training: to learn or train a model
  - Testing: to validate the model



Kitten A Kitten B Kitten C

Training database



1 2 3 4

Testing database

Cross Validation

- It is used for
  - Performance evaluation: Evaluate the performance of a classifier using the given data
  - Model Selection: Compare the performance of two or more algorithms (DT classifier and neural network) to determine the best algorithm for the given data
  - Tuning model parameters: Compare the performance of two variants of a parametric model

Type of Cross Validation

- Resubstitution Validation
- Hold-Out Validation
- K-Fold Cross-Validation
- Leave-One-Out Cross-Validation
- Repeated K-Fold Cross-Validation

Type of Cross Validation…

- Resubstitution Validation
  - All the available data is used for training and the same data is used for testing
    - Does not provide any information about generalizability

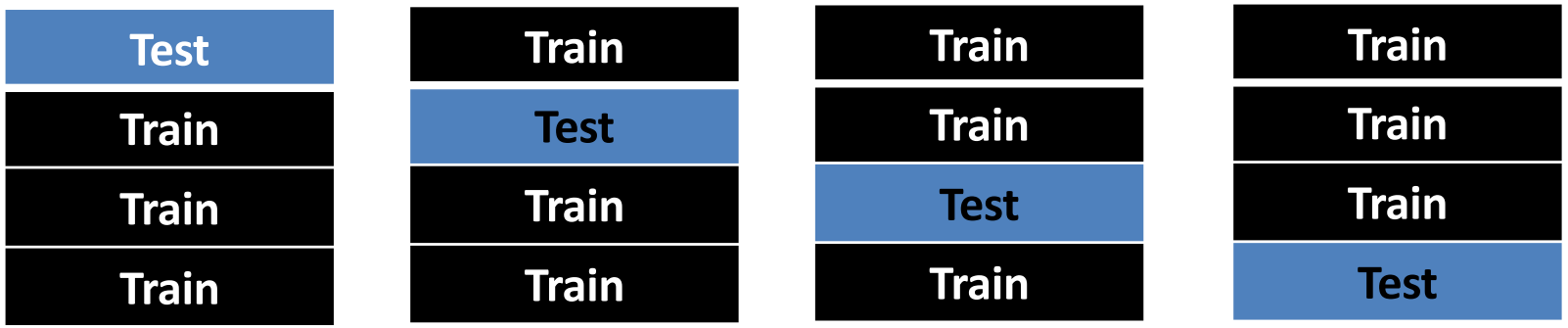
Type of Cross Validation…

- Hold-Out Validation
  - The database is partitioned into two non-overlapping parts, one for training and other for testing
    - The results depend a lot on the partition, may be skewed if the test set is too easy or too difficult

Type of Cross Validation…

- K-Fold Cross-Validation
  - Data is partitioned into k equal folds (partitions). k-1 folds are used for training and 1-fold for testing
  - The procedure is repeated k times

- Across multiple folds, report:
  - Average error or accuracy
  - Standard deviation or variance



4-fold cross validation

Type of Cross Validation…

- Repeated K-Fold Cross-Validation
  - Repeat k-fold cross validation multiple times
- Leave-One-Out Cross-Validation
  - Special case of k-fold cross validation where k=number of instances in the data
  - Testing is performed on a single instance and the remaining are used for training

- Across multiple folds, report:
  - Average error or accuracy
  - Standard deviation or variance

Comparing Cross-Validation Methods

Validation Method	Advantages	Disadvantages
Resubstitution	Simple	Overfitting
Hold-out validation	Independent training and testing sets	Reduced data for training and testing
K-fold cross validation	Accurate performance estimation	Small sample for performance estimation, underestimated performance variance or overestimated degree of freedom for comparison
Leave-one-out cross validation	Unbiased performance estimation	Very large variance
Repeated k-fold cross-validation	Large number of performance estimates	Overlapped training and test data between each round, underestimated performance variance or overestimated degree of freedom for comparison